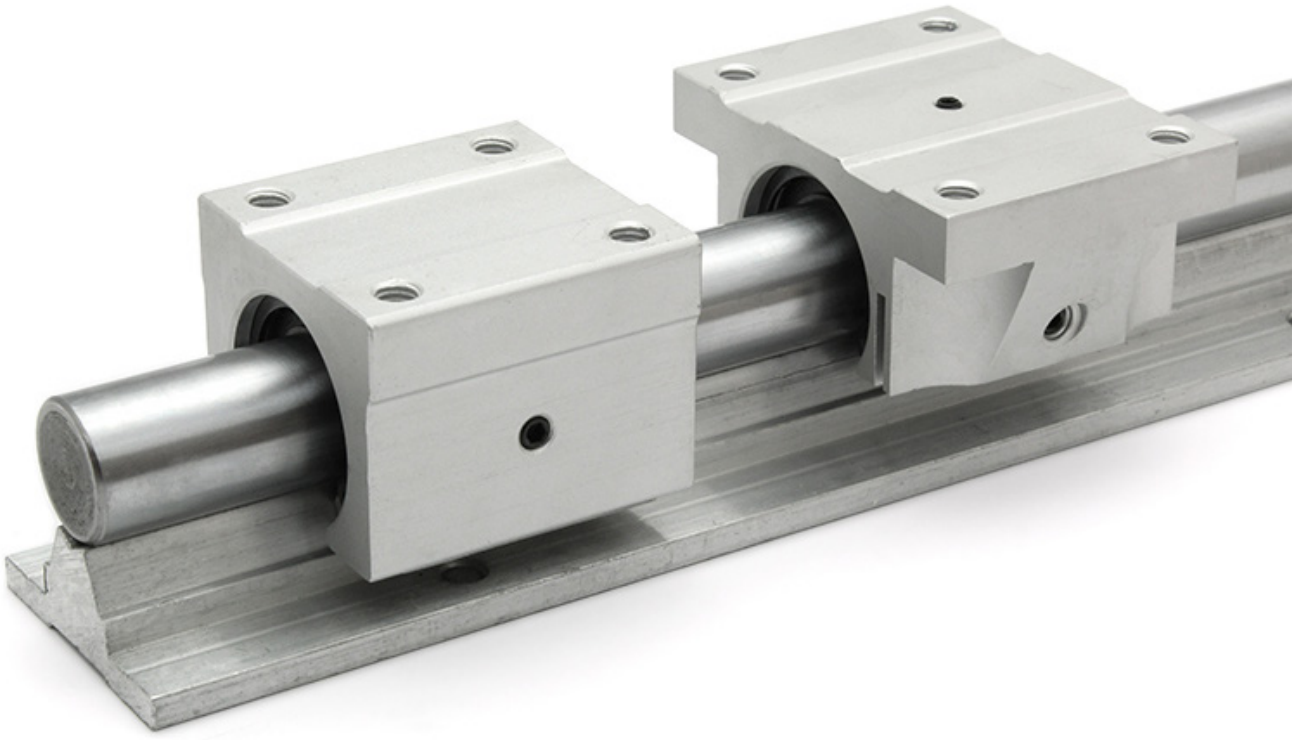
DATASHEET



Supported Rails

and Suitable Bearings

Overview:

Rails

- **SBS**
- **TBS**

Linear Bearings

with housing:

- **SBR···UU**
- **SBR···LUU**
- **TBR···UU**

without housing:

- **LM···UU**

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Version: 2024_03_15

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- Supported Rails in combination with Aluminium profiles
- Which aluminum profiles fit?
- SBS or TBS?
- Which bearings fit?



Supported Rail SBS — **Pages 5-6**
- Technical Data
- Ordering Information with standard length



Supported Rail TBS — **Pages 7-8**
- Technical Data
- Ordering Information with standard length



Linear Bearing SBR···UU for Supported Rail — **Page 9**
- Technical Data
- Ordering Information



Linear Bearing SBR···LUU for Supported Rail — **Page 10**
- Technical Data
- Ordering Information

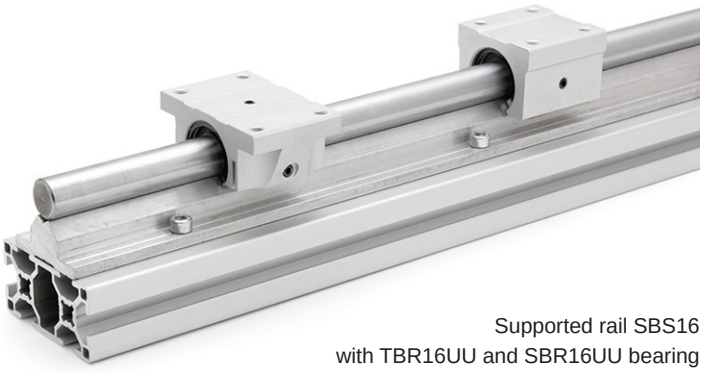


Linear Bearing TBR···UU for Supported Rail — **Page 11**
- Technical Data
- Ordering Information

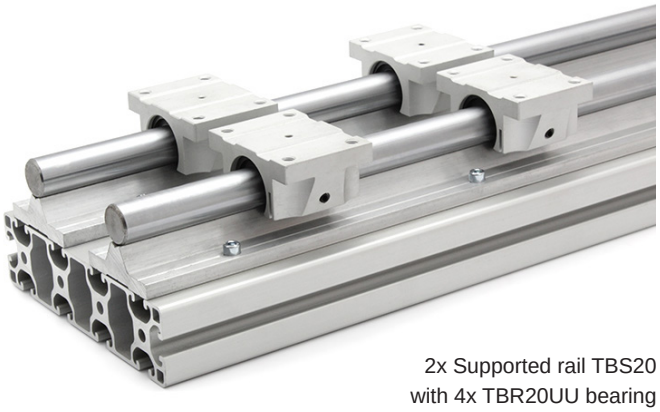


Linear Bearing LM···UU for Supported Rail — **Pages 12-13**
- Technical Data
- Ordering Information

Information about Supported Rails



Supported rail SBS16
with TBR16UU and SBR16UU bearing
on a B-Typ Nut 8 30x60 profile



2x Supported rail TBS20
with 4x TBR20UU bearing
on a I-Typ Nut 8 40x160 profile

Supported Rails in Combination with Alumium Profiles

Do you want to mount your supported rail on an aluminum profile? Then note the distance between the mounting holes (C). This must correspond to the distance between the grooves in the aluminum profile. The following table gives you an overview of possible combinations.

Which Aluminium Profiles Fit?

Bezeichnung / *Type*	**Abstand der Bohrungs-befestigungen C /** *Distance between the hole fixings C*	**Geeignete Aluminiumprofile /** *Suitable aluminium profiles*	
		Profilraster / *Profile grid*	**Maße /** *Dimensions*
SBS16	30	30	I-Typ Nut / *Goove* 6 30x60 I-Typ Nut/ *Goove* 6 30x120 I-Typ Nut/ *Goove* 6 60x60 B-Typ Nut/ *Goove* 8 30x60
SBS20	30	30	I-Typ Nut / *Goove* 6 30x60 I-Typ Nut / *Goove* 6 30x120 I-Typ Nut / *Goove* 6 60x60 B-Typ Nut / *Goove* 8 30x60
SBS30	40	40	I-Typ Nut / *Goove* 8 40x80 I-Typ Nut / *Goove* 8 40x120 I-Typ Nut / *Goove* 8 40x160 I-Typ Nut / *Goove* 8 80x80
SBS35	45	45	B-Typ Nut / *Goove* 10 45x90 B-Typ Nut / *Goove* 10 90x90
TBS20	40	40	I-Typ Nut / *Goove* 8 40x80 I-Typ Nut / *Goove* 8 40x120 I-Typ Nut / *Goove* 8 40x160 I-Typ Nut / *Goove* 8 80x80
TBS25	45	45	B-Typ Nut / *Goove* 10 45x90 B-Typ Nut / *Goove* 10 90x90

Information about Supported Rails



SBS



TBS

SBS or TBS ?

The SBS and TBS rails differ in the dimensions of the aluminum substructure.
The main difference is the width - SBS rails are narrower than the TBS rails.
The heights differ by a few millimeters.
TBR and SBR bearings can be combined with both TBS and SBS supported rails.

Which Bearings fit?

The following linear bearings can be combined with both SBS and TBS supported rails. All technical data and article numbers can be found on the following pages.



SBR···UU



SBR···LUU



TBR···UU

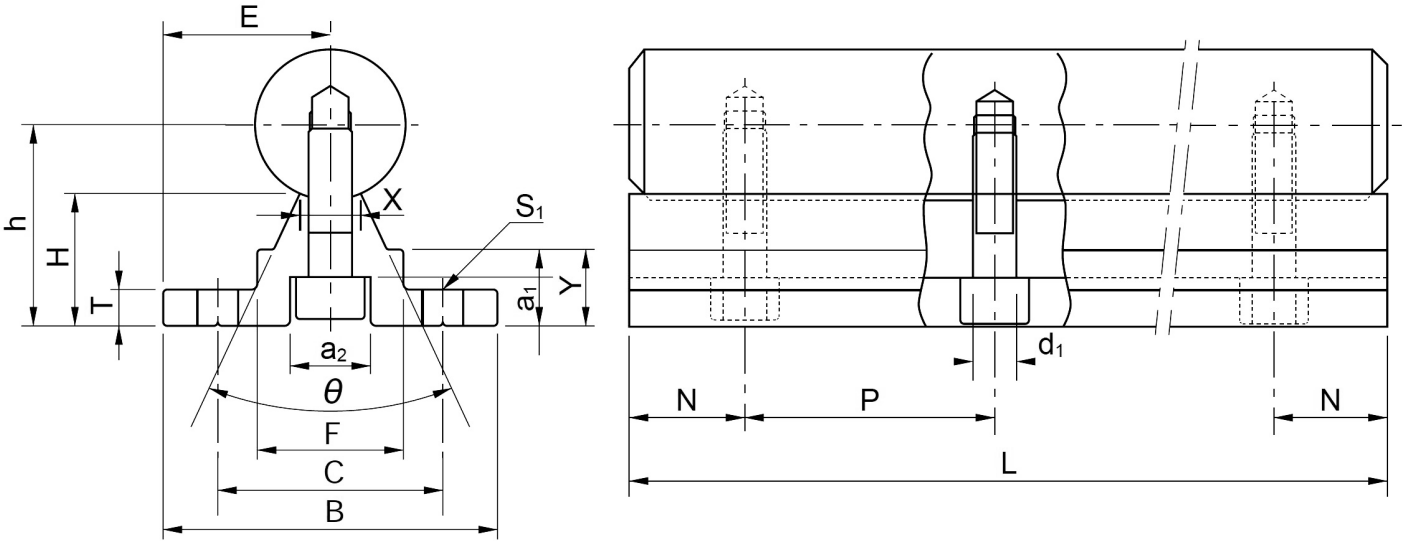


LM···UU-OP

Supported Rail SBS



Material of the shafts: GCr15 High carbon chromium steel
Hardness: HRC62 ±2
Roughness: Ra0.8
Straightness: better than 50 µm / 300 mm
Tolerance: g6
Material of the base: Aluminium



Bezeichnung / *Type*	Wellendurchmesser / *Shaft Diameter*	Maße / *Dimensions* (mm)														Gewicht / *Weight* (g/m)
		E	h	B	H	T	F	X	Y	C	θ	S_1	a_1	a_2	d_1	
SBS12	Ø12	15,8	18,5	31,6	13	4	12,5	6	8	22	65°	Ø4,5	5	8	4,5	1450
SBS16	Ø16	20	25	40	17,8	5	18,5	8	11,7	30	60°	Ø5,5	6	9,5	5,5	2560
SBS20	Ø20	22,5	27	45	17,7	5	19	8	10	30	50°	Ø5,5	6,5	11	6,6	3500
SBS25	Ø25	27,5	33	55	21	6	21,5	8	12	35	50°	Ø6,6	6,5	11	6,6	5300
SBS30	Ø30	30	37	60	22,8	7	26,5	10,3	13	40	50°	Ø6,6	8,5	14	9	7380
SBS35	Ø35	32,5	43	65	26,5	8	28	13	15,5	45	50°	Ø9,0	8,5	14	9	9680
SBS40	Ø40	37,5	48	75	29,4	9	38	16	17	55	50°	Ø9,0	8,5	14	9	12690
SBS50	Ø50	47,5	62	95	38,8	11	45	20	21	70	50°	Ø11	12,5	19	11	20460

Supported Rail SBS

We offer other lengths as special designs.
Contact us via info@dold-mechatronik.de.

L Länge / *Length* (mm)	Bezeichnung / *Type*							
	SBS12 (mm)	SBS16 (mm)	SBS20 (mm)	SBS25 (mm)	SBS30 (mm)	SBS35 (mm)	SBS40 (mm)	SBS50 (mm)
100	P=50 , N=25							
Art. Nr.	34917	*	*	*	*	*	*	*
150	P=100 , N=25							
Art. Nr.	30001	*	*	*	*	*	*	*
200	P=100 , N=50	P=150 , N=25	P=150 , N=25					
Art. Nr.	30018	12014	12137	*	*	*	*	*
250	P=100 , N=25	P=150 , N=50	P=150					
Art. Nr.	30025	12564	SBS20-250	*	*	*	*	*
300	P=100 , N=50	P=150 , N=75	P=150 , N=75					
Art. Nr.	30032	12021	12144	*	*	*	*	*
350	P=100 , N=25	P=150 , N=25	P=150					
Art. Nr.	30049	12570	SBS20-350	*	*	*	*	*
400	P=100 , N=50	P=150 , N=50	P=150 , N=50		P=200 , N=100			
Art. Nr.	30056	12038	12151	*	12762	*	*	*
450	P=100 , N=25	P=150 , N=75	P=150					
Art. Nr.	30063	12625	SBS20-450	*	*	*	*	*
500	P=100 , N=50	P=150 , N=25	P=150 , N=25	P=200 , N=50	P=200 , N=50			
Art. Nr.	30070	12045	12168	12281	12694	*	*	*
600	P=100 , N=50	P=150 , N=75	P=150 , N=75	P=200 , N=100	P=200 , N=100			
Art. Nr.	30087	12052	12175	12656	12700	*	*	*
700	P=100 , N=50	P=150 , N=50	P=150					
Art. Nr.	30094	12557	12632	*	*	*	*	*
800	P=100 , N=50	P=150 , N=25	P=150 , N=25	P=200 , N=100	P=200 , N=100			
Art. Nr.	30100	12069	12182	12298	12717	*	*	*
1000	P=100 , N=50	P=150 , N=50	P=150 , N=50	P=200 , N=100	P=200 , N=100		P=200 , N=100	
Art. Nr.	30117	12076	12199	12304	12380	*	12458	*
1200	P=100 , N=50	P=150 , N=75	P=150 , N=75	P=200 , N=100	P=200 , N=100			
Art. Nr.	30124	12083	12205	12311	12687	*	*	*
1500	P=100 , N=50	P=150 , N=75	P=150 , N=75	P=200 , N=50	P=200 , N=50		P=200 , N=50	
Art. Nr.	30131	12090	12212	12328	12397	*	12465	*
2000	P=100 , N=50	P=150 , N=25	P=150 , N=25	P=200 , N=100	P=200 , N=100		P=200 , N=100	
Art. Nr.	30148	12113	12229	12335	12403	*	12472	*
2500		P=150 , N=50	P=150 , N=50	P=200 , N=50	P=200 , N=50		P=200 , N=50	
Art. Nr.	*	12120	12236	12342	12410	*	12489	*
3000			P=150 , N=75	P=200 , N=100	P=200 , N=100		P=200 , N=100	
Art. Nr.	*	*	12243	12359	12427	*	12496	*
3200			P=150 , N=25	P=200 , N=100				
Art. Nr.	*	*	12250	12663	*	*	*	*
3500			P=150 , N=25	P=200 , N=50	P=200 , N=50			
Art. Nr.	*	*	12267	12366	12434	*	*	*
4000			P=150 , N=50	P=200	P=200 , N=100		P=200 , N=100	
Art. Nr.	*	*	12274	12373	12441	*	12502	*

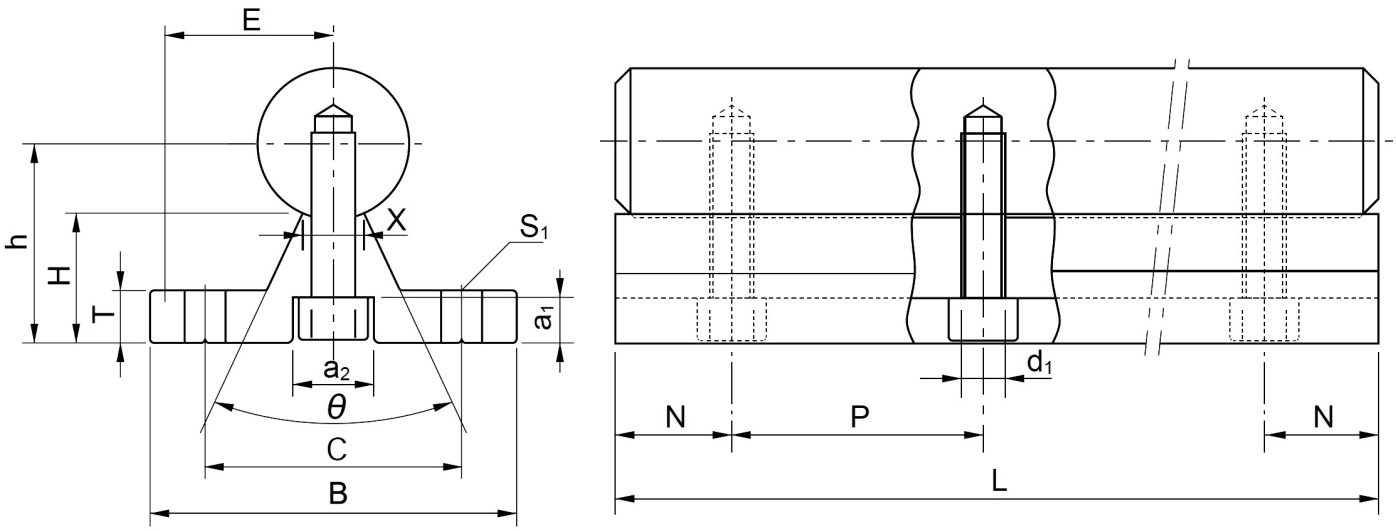
* on demand

P=hole spacing, N=distance from the first hole to the edge

Supported Rail TBS



Material of the shaft: GCr15 High carbon chromium steel
Hardness: HRC62+/-2
Roughness: Ra0.8
Straightness: better than 50 µm / 300 mm
Tolerance: g6
Material of the base: Aluminium



Bezeichnung / *Type*	Wellendurchmesser / *Shaft Diameter*	Maße / *Dimensions* (mm)												Gewicht / *Weight* (g/m)
		E	h	B	H	T	X	C	θ	S_1	a_1	a_2	d_1	
TBS16	Ø16	25	22	50	14,84	6	8	37	60°	5,5	6	9,5	5,5	2660
TBS20	Ø20	27,5	29	55	19,64	8	8	40	50°	5,5	6,5	11	6,6	4230
TBS25	Ø25	32,5	32	65	20,00	10	8	45	50°	6,6	6,5	11	6,6	5850
TBS30	Ø30	37,5	36,5	75	22,28	12	10,3	55	50°	6,6	8,5	14	9	8280

Supported Rail TBS

We offer other lengths as special designs.
Contact us via info@dold-mechatronik.de.

L Länge / *Length* (mm)	Bezeichnung / *Type*			
	TBS16 (mm)	TBS20 (mm)	TBS25 (mm)	TBS30 (mm)
100				
Art. Nr.	*	*	*	*
150				
Art. Nr.	*	*	*	*
200	P=150 , N=25	P=150 , N=25		
Art. Nr.	**11567**	**11673**	*	*
250	P=150 , N=50	P=150 , N=50		
Art. Nr.	**12588**	**12779**	*	*
300	P=150 , N=75	P=150		
Art. Nr.	**11574**	**11680**	*	*
350	P=150 , N=25	P=150		
Art. Nr.	**12595**	**12786**	*	*
400	P=150 , N=50	P=150 , N=50		
Art. Nr.	**11581**	**11697**	*	*
450		P=150		
Art. Nr.	*	**12793**	*	*
500	P=150 , N=25	P=150 , N=25	P=200 , N=50	P=200 , N=50
Art. Nr.	**11598**	**11703**	**11826**	**12731**
600	P=150 , N=75	P=150 , N=75	P=200 , N=100	P=200 , N=100
Art. Nr.	**11604**	**11710**	**12670**	**12748**
700	P=150 , N=50	P=150		
Art. Nr.	**12649**	**12809**	*	*
800	P=150 , N=25	P=150 , N=25	P=200 , N=100	P=200 , N=100
Art. Nr.	**11611**	**11727**	**11833**	**12755**
1000	P=150 , N=50	P=150 , N=50	P=200 , N=100	P=200 , N=100
Art. Nr.	**11628**	**11734**	**11840**	**11932**
1200	P=150 , N=75	P=150 , N=75	P=200 , N=100	P=200 , N=100
Art. Nr.	**11635**	**11741**	**11857**	**12724**
1500	P=150 , N=75	P=150 , N=75	P=200 , N=50	P=200 , N=50
Art. Nr.	**11642**	**11758**	**11864**	**11949**
2000	P=150 , N=25	P=150 , N=25	P=200 , N=100	P=200 , N=100
Art. Nr.	**11659**	**11765**	**11871**	**11956**
2500	P=150 , N=50	P=150 , N=50	P=200 , N=50	P=200 , N=50
Art. Nr.	**11666**	**11772**	**11888**	**11963**
3000		P=150 , N=75	P=200 , N=100	P=200 , N=100
Art. Nr.	*	**11789**	**11895**	**11970**
3200		P=150 , N=25	P=200 , N=100	P=200 , N=100
Art. Nr.	*	**11796**	**11901**	**12618**
3500		P=150 , N=25	P=200 , N=50	P=200 , N=50
Art. Nr.	*	**11802**	**11918**	**11994**
4000		P=150 , N=50	P=200 , N=100	P=200 , N=100
Art. Nr.	*	**11819**	**11925**	**12007**

* on demand

P=hole spacing, N=distance from the first hole to the edge

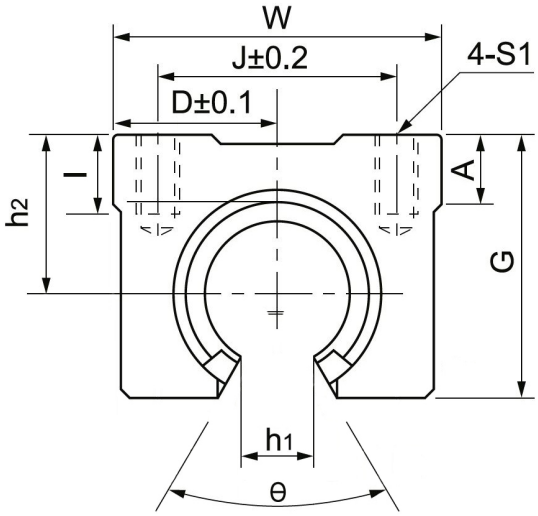
Linear Bearing with Housing - open - standard

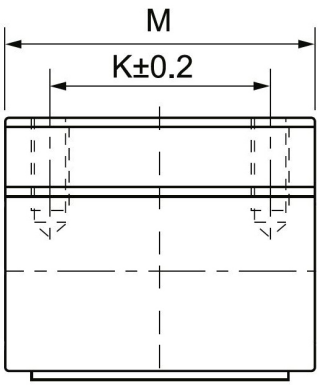
SBR···UU



Bezeichnung / *Type*	Linearlager / *Linear Bearing*	Tragzahl / *Basic load rating*	
		Dynamisch C N *Dynamic C N*	Statisch Co N *Static Co N*
SBR12UU	LM12UUOP	500	770
SBR16UU	LM16UUOP	770	1170
SBR20UU	LM20UUOP	860	1370
SBR25UU	LM25UUOP	980	1560
SBR30UU	LM30UUOP	1560	2740
SBR35UU	LM35UUOP	1660	3130
SBR40UU	LM40UUOP	2150	4010
SBR50UU	LM50UUOP	3820	7930

Linear bearing with aluminum housing for supported shaft guides (supported rails).
This linear bearing can be combined with both SBS and TBS supported rails.
UU = Dirt wipers on both sides.





Art.Nr. / *Item No.*	Bezeichnung / *Type*	Wellendurchmesser / *Shaft Diameter*	Maße / *Dimensions* (mm)												Gewicht / *Weight* (g)
			M	W	G	A	I	h_1	h_2	S_1	D	J	K	θ	
34900	SBR12UU	Ø12	39	40	27,6	8	10	8,5	17	M5x10	20	28	26	80°	150
11376	SBR16UU	Ø16	45	45	33	9	12	10	20	M5x12	22,5	32	30	80°	150
11383	SBR20UU	Ø20	50	48	39	11	12	10	23	M6x12	24	35	35	60°	200
11390	SBR25UU	Ø25	65	60	47	14	12	11,5	27	M6x12	30	40	40	50°	450
11406	SBR30UU	Ø30	70	70	56	15	18	14	33	M8x18	35	50	50	50°	630
*	SBR35UU	Ø35	80	80	63	18	18	16	37	M8x18	40	55	55	50°	920
11413	SBR40UU	Ø40	90	90	72	20	20	19	42	M10x20	45	65	65	50°	1330
11420	SBR50UU	Ø50	110	120	92	25	20	23	53	M10x20	60	94	80	50°	3000

* on demand
** without side adjustment screw

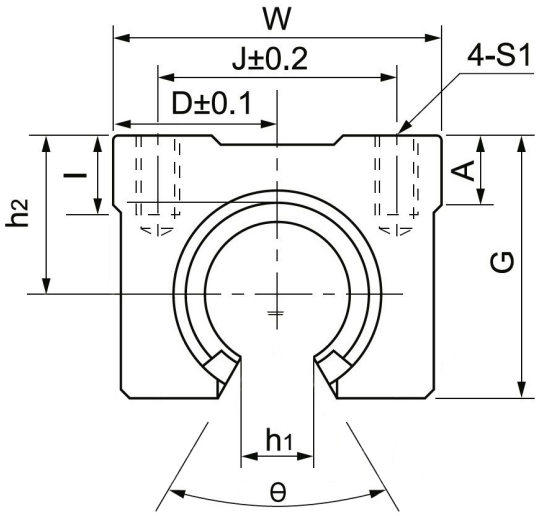
Linear Bearing with Housing - open - long

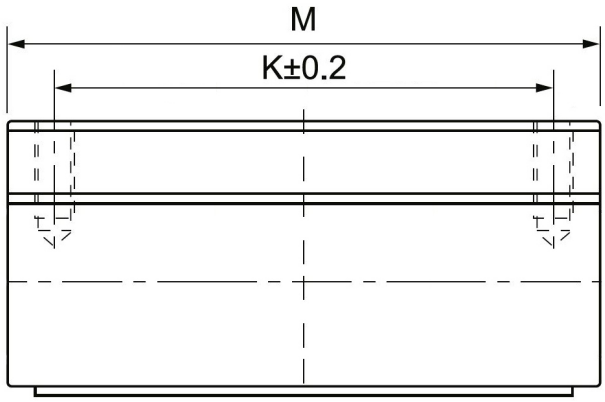
SBR···LUU



Bezeichnung / *Type*	Linearlager / *Linear Bearing*	Tragzahl / *Basic load rating*	
		Dynamisch C N *Dynamic C N*	Statisch Co N *Static Co N*
SBR16LUU	2xLM16UUOP	1548	2360
SBR20LUU	2xLM20UUOP	1764	2740
SBR25LUU	2xLM25UUOP	1960	3140
SBR30LUU	2xLM30UUOP	3140	5480
SBR40LUU	2xLM40UUOP	4320	8040

Linear bearing with aluminum housing for supported shaft guides (supported rails).
This linear bearing can be combined with both SBS and TBS supported rails.
The linear bearing combines two recirculating ball bushings in one housing.
UU = Dirt wipers on both sides.





Art.Nr. / *Item No.*	Bezeichnung / *Type*	Wellendurchmesser / *Shaft Diameter*	Maße / *Dimensions* (mm)												Gewicht / *Weight* (g)
			M	W	G	A	I	h_1	h_2	S_1	D	J	K	θ	
SBR12LUU		Ø12	70	40	27	8	10	NV	17	M5x10	20	28	46	80°	170
11437	SBR16LUU	Ø16	85	45	33	9	12	10	20	M5x12	22,5	32	60	80°	300
11444	SBR20LUU	Ø20	96	48	39	11	12	10	23	M6x12	24	35	70	60°	400
11451	SBR25LUU	Ø25	130	60	47	14	12	11,5	27	M6x12	30	40	100	50°	900
11468	SBR30LUU	Ø30	140	70	56	15	18	14	33	M8x18	35	50	110	50°	1260
11475	SBR40LUU	Ø40	175	90	72	20	20	19	42	M10x20	45	65	140	50°	2660

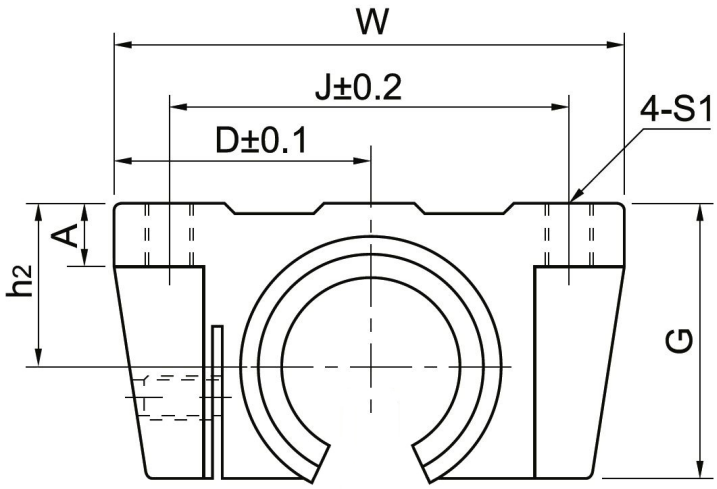
Linear Bearing with Housing - open - adjustable

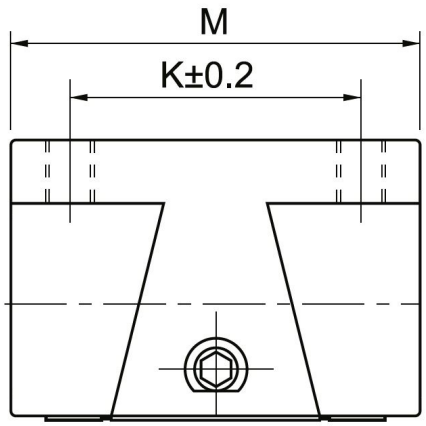
TBR···UU



Bezeichnung / *Type*	Linearlager / *Linear Bearing*	Tragzahl / *Basic load rating*	
		Dynamisch C N *Dynamic C N*	Statisch Co N *Static Co N*
TBR16UU	LM16UUOP	392	490
TBR20UU	LM20UUOP	784	1176
TBR25UU	LM25UUOP	1568	2352
TBR30UU	LM30UUOP	1764	2940

Linear bearing with aluminum housing for supported shaft guides (supported rails).
This linear bearing can be combined with both SBS and TBS supported rails.
UU = Dirt wipers on both sides.





Art.Nr. / *Item No.*	Bezeichnung / *Type*	Wellendurchmesser / *Shaft Diameter*	Maße / *Dimensions* (mm)									Gewicht / *Weight* (g)
			M	W	G	A	h_2	D	S_1	J	K	
11482	TBR16UU	Ø16	42	62	26	8	18	31	M5	50	30	180
11499	TBR20UU	Ø20	51	68	31	10	21	34	M6	54	37	300
11505	TBR25UU	Ø25	65	82	41	12	28	41	M8	65	50	600
11512	TBR30UU	Ø30	75	91	48	12	33,5	45,5	M8	75	60	900

Linear Bearing without Housing - open - standard

LM···UU-OP

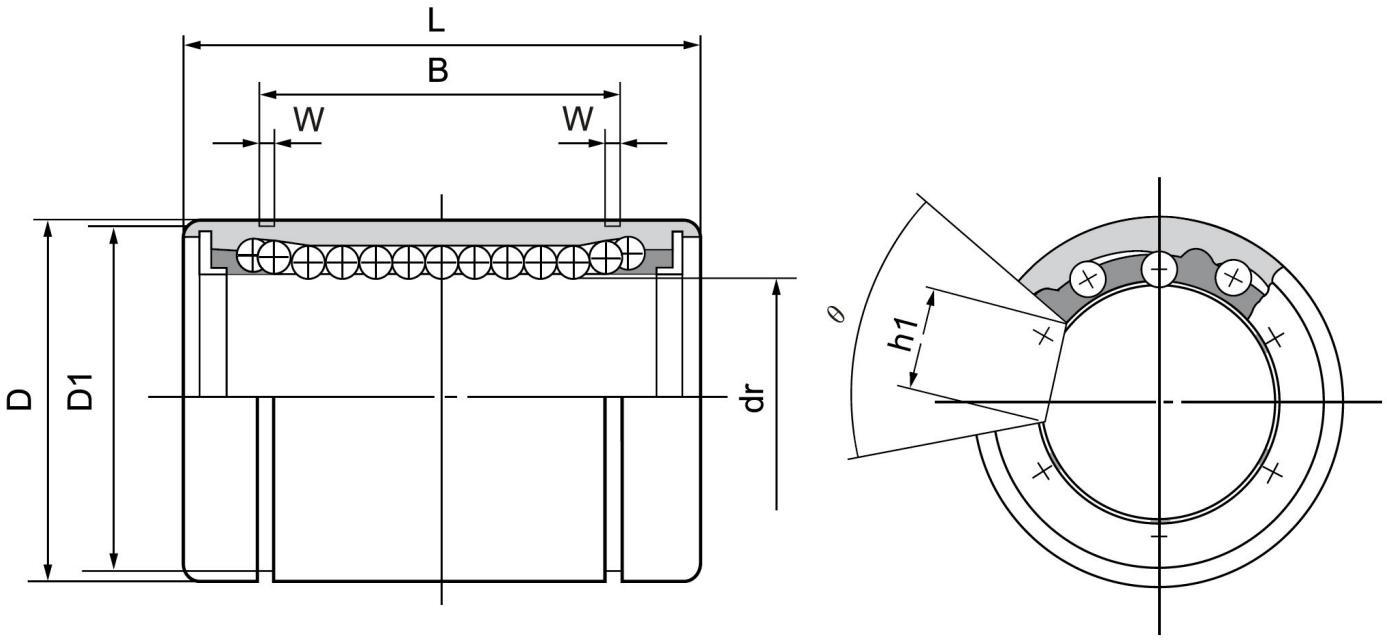


LM···UU-OP LBearing according to the metric-European standard.
UU = Dirt wipers on both sides.
OP = open

Bezeichnung / *Type*	**Wellendurch-messer /** *Shaft Diameter*	**Anzahl Kugelumläufe /** *Ball circuits*	**Exzentrizität /** *Eccentricity* **(µm)**		**Radialspiel max. /** *Radial clearance max.* **(µm)**	**Tragzahl /** *Basic load rating*		**Gewicht /** *Weight* **(g)**
			Precision	*High*		**Dynamisch C N** *Dynamic C N*	**Statisch Co N** *Static Co N*	
LM10UU-OP	Ø10	4	8	12	-4	372	549	29
LM12UU-OP	Ø12	4	8	12	-4	510	784	31
LM13UU-OP	Ø13	4	8	12	-4	510	784	42
LM16UU-OP	Ø16	5	8	12	-6	774	1180	68
LM20UU-OP	Ø20	5	10	15	-6	882	1370	85
LM25UU-OP	Ø25	6	10	15	-6	980	1570	216
LM30UU-OP	Ø30	6	10	15	-8	1570	2740	245
LM35UU-OP	Ø35	6	12	20	-8	1670	3140	384
LM40UU-OP	Ø40	6	12	20	-10	2160	4020	579
LM50UU-OP	Ø50	6	12	20	-13	3820	7940	1560
LM60UU-OP	Ø60	6	17	25	-13	4700	10000	1820
LM80UU-OP	Ø80	6	17	25	-20	7350	16000	4300
LM100UU-OP	Ø100	6	20	30	-20	14100	34800	8540
LM120UU-OP	Ø120	8	20	30	-25	16400	40000	14900
LM150UU-OP	Ø150	8	25	40	-25	21100	54300	20150

Linear Bearing without Housing - open - standard

LM···UU-OP



Art.Nr. / *Item No.*	Bezeichnung / *Type*	dr			Maße und Toleranzen / *Dimensions and tolerances*									
		mm	Toleranz / *Tolerance* µm		D		L		B		W	D_1	h_1	θ
			Precision	*High*	*mm*	*Tol. µm*	*mm*	*Tol. µm*	*mm*	*Tol. µm*	*mm*	*mm*	*mm*	
10140	LM10UU-OP	10	0 / -6	0 / -9	19	0 / -13	29	0 / -200	22	0 / -200	1,3	18	6,8	80°
10157	LM12UU-OP	12	0 / -6	0 / -9	21	0 / -13	30	0 / -200	23	0 / -200	1,3	20	8	80°
LM13UU-OP		13	0 / -6	0 / -9	23	0 / -13	32	0 / -200	23	0 / -200	1,3	22	9	80°
10164	LM16UU-OP	16	0 / -6	0 / -9	28	0 / -13	37	0 / -200	26,5	0 / -200	1,6	27	11	80°
10171	LM20UU-OP	20	0 / -7	0 / -10	32	0 / -16	42	0 / -200	30,5	0 / -200	1,6	30,5	11	60°
10188	LM25UU-OP	25	0 / -7	0 / -10	40	0 / -16	59	0 / -300	41	0 / -300	1,85	38	12	50°
10195	LM30UU-OP	30	0 / -7	0 / -10	45	0 / -16	64	0 / -300	44,5	0 / -300	1,85	43	15	50°
LM35UU-OP		35	0 / -8	0 / -12	52	0 / -19	70	0 / -300	49,5	0 / -300	2,1	49	17	50°
10201	LM40UU-OP	40	0 / -8	0 / -12	60	0 / -19	80	0 / -300	60,5	0 / -300	2,1	57	20	50°
LM50UU-OP		50	0 / -8	0 / -12	80	0 / -19	100	0 / -300	74	0 / -300	2,6	76,5	25	50°
LM60UU-OP		60	0 / -9	0 / -15	90	0 / -22	110	0 / -300	85	0 / -300	3,15	86,5	30	50°
*	LM80UU-OP	80	0 / -9	0 / -15	120	0 / -22	140	0 / -400	105,5	0 / -400	4,15	116	40	50°
*	LM100UU-OP	100	0 / -10	0 / -20	150	0 / -25	175	0 / -400	125,5	0 / -400	4,15	145	50	50°
*	LM120UU-OP	120	0 / -10	0 / -20	180	0 / -25	200	0 / -400	158,6	0 / -400	4,15	175	85	80°
*	LM150UU-OP	150	0 / -13	0 / -25	210	0 / -29	240	0 / -400	170,6	0 / -400	5,15	204	105	80°

* on demand